BEAI

Biochemical Engineering AI

A summary of the platform, the milestones, and the measurements that were written down.

Compiled from repository documents dated 2026-10-11 through 2026-10-13, the milestone status notes, and the 11 October 2026 prompt pack. This PDF does not add a measurement, a citation, or a benchmark.

> **Research software.** BEAI is not a clinical tool. It does not give medical advice. A docking score is not an affinity. A model output is not an experimental measurement. A short simulation is not evidence of stability, binding, or efficacy.

What BEAI is

BEAI (Biochemical Engineering AI) is a modular research platform for protein analysis, compound analysis, computational drug discovery, and therapeutic-safety evidence review. The prompt pack and `AGENTS.md` use that definition.

The app that was built is local. A FastAPI service stores analyses in SQLite. A Vite and React interface shows a structure in a 3Dmol.js viewer and downloads HTML, Markdown, and JSON reports. The README quick start serves the API on port 8731 and the interface on port 8732. Uploads are PDB or mmCIF. Sequences can be pasted or imported from UniProt. Pockets, chain comparison, and docking run as jobs. The Model Laboratory lists registered models and a live RAM report. Weights are not loaded at startup.

Every analysis returns one AnalysisResult envelope: inputs, methods, measurements with an origin and a unit, evidence with a category, limitations, and a reproducibility block. The origins are experimental_record, database_reported, calculated, ml_prediction, and user_supplied. Evidence categories are established, observational, computational_prediction, hypothesis, and unclassified. An absent value is null plus a warning.

Goals

The prompt pack states the priorities in this order. The same sentence says to integrate existing scientific tools and pretrained models before considering custom training.

1	Scientific correctness
2	Working software
3	Reproducibility
4	Modularity
5	Honest uncertainty

Source: internal/beai-cursor-prompt-pack.pdf, rule 00, Mission. The roadmap repeats the build order: foundation, then protein, compounds, docking, safety evidence, then six model-and-simulation sessions.

Milestones

Each note below is the pack’s aim for that chat, then what the status notes record as run. A pass count is the full pytest suite at the end of that work, from the status note named with it.

1 · Foundation

The pack asks for an offline path from a PDB or mmCIF upload to measurements computed from that file, a 3D view, a saved analysis that survives a restart, and HTML, Markdown, and JSON reports. Branch `m1-foundation` records that path: structure overview, SQLite, 3Dmol.js, and reports. Unbuilt areas were Planned pages with no actions. pytest 34 passed, 0 failed. Vitest 7 passed, 0 failed. The Playwright smoke test passed 1, failed 0. At that health check, RDKit, mkdssp, Vina, OpenMM, PyTorch, and GPU were unavailable, with reasons. Source: docs/milestone-1-status.md.

2 · Protein

The pack asks for sequence statistics, DSSP or Biotite P-SEA, SASA, contacts, hydrogen-bond estimates, superposition, a baseline pocket detector with a held-out benchmark, and RCSB and UniProt import with provenance. Branch `m2-protein` records that work, with an in-process job runner. mkdssp 4.2.2 was on PATH, so 1UBQ secondary structure used DSSP. fpocket was not installed. Held-out pocket success was 3 of 5 at the pre-registered 4.0 Å cutoff. pytest 55 passed, 1 deselected (network), 0 failed. Vitest 7 passed, 0 failed. Playwright was not re-run. Source: docs/remaining-plan-status.md.

3 · Compounds

The pack asks for RDKit sanitization, descriptors named by their RDKit functions, Tanimoto similarity, and PubChem and ChEMBL records kept distinct from calculated values. Aspirin, caffeine, and ethanol matched the contract checks: C9H8O4 and 180.16, C8H10N4O2 and 194.19, C2H6O and 46.07, using RDKit 2025.09.1 `Descriptors.MolWt` rounded to 2 decimals. ChEMBL activity import is capped at 100 rows. The molecule JSON had no licence field, so none was invented. pytest 66 passed, 1 deselected, 0 failed. Vitest 7 passed. Playwright was not re-run. Branch `m3-compounds`.

4 · Docking

The pack asks for a real AutoDock Vina run when the tool is installed, HTTP 503 when it is not, and a redocking table that includes failures. No simulated pose. This milestone recorded a Vina 1.2.5 run on 3PTB / benzamidine. Parser scores from the captured PDBQT are −6.073, −5.062, and −4.839 kcal/mol. A second seed-1 run differed by 0.006 kcal/mol, inside the 0.05 kcal/mol window written down for that spread. The 5-to-10 complex heavy-atom RMSD benchmark was not run in this milestone. Meeko 0.8.0 was installed because 0.6.1 does not support Python 3.12. pytest 71 passed, 1 deselected, 0 failed. A frontend unit-test count was not recorded. Branch `m4-docking`.

5 · Safety evidence

The pack asks for source-linked safety evidence with mandatory categories, and for no medical advice. The validation file says this module is not a clinical model and has no accuracy target. FAERS stays observational. Similarity off-targets stay computational_prediction and “unverified”. Literature stays unclassified until a person sets a category. At the end of the milestone, Open Targets, DailyMed fallback, and Europe PMC were not built. pytest 78 passed, 1 deselected (network), 0 failed. A frontend unit-test count was not recorded. Branch `m5-safety`.

6A · Registry

The pack asks for a model registry before any weights are used, a live hardware report, and load and unload that do not run at startup. Branch `m6a-registry` stores name, version, task, source, licence, dependencies, hardware, formats, an optional benchmark, and known limitations. A missing dependency is disabled (HTTP 503). A declared requirement the machine does not meet is incompatible (HTTP 422). The registry-only loader does not execute a weight file. pytest 88 passed, 1 deselected, 0 failed. Vitest 9 passed, 0 failed.

6B · Embeddings

The pack asks for one small protein language model, loaded on demand, and a rank correlation between embedding cosine similarity and alignment identity. ESM-2 8M (`facebook/esm2_t6_8M_UR50D`) is registered. The card licence recorded in the status note is MIT. `max_position_embeddings` is 1026. Sequences over that limit are refused, not chunked. Spearman on eight local chains (28 pairs) is 0.6990131578947367, with a bootstrap interval from 0.3788637829986782 to 0.8820213682475236. The validation file says this is rank agreement with sequence identity, not a functional or structural result. pytest 94 passed, 2 deselected (network and model), 0 failed. The opt-in model test passed 1. Vitest 9 passed. Branch `m6b-embeddings`.

6C · Properties

The pack asks for a public solubility set with a checksum, a scaffold split saved before fitting, simple baselines beside any neural model, and RMSE, MAE, and R-squared with bootstrap intervals. The Delaney table has 1,128 data rows. The split is 902 train and 226 test, 0 rows dropped, seed 20261013. Ridge alpha was 1.0 and was not tuned. The two-layer graph net (hidden size 32, 20 epochs, Adam learning rate 0.01) was not tuned on the test split. Its R-squared is negative: worse than the mean of the test labels. pytest 98 passed, 3 deselected, 0 failed. The held-out metrics test passed 1. Vitest 9 passed. Branch `m6c-properties`.

6D · Simulation

The pack asks for PDBFixer preparation, energy minimization labeled as not molecular dynamics, and a hard-capped short simulation. OpenMM 8.6.1 reported platforms Reference and CPU. CUDA, OpenCL, and Metal were not in that list. A 20-iteration minimization probe on 1UBQ lowered the potential energy. The pre-registered NVE relative-drift bound of 0.02 was not stable. Langevin mean kinetic temperature was inside 150 K to 450 K. The same-seed total-energy gap missed 1.0 kJ/mol. The session status says checkpoint resume was not implemented yet. pytest 103 passed, 3 deselected, 0 failed. Vitest 9 passed. Branch `m6d-simulation`.

6E · Generation

The pack treats generation as optional and tightly scoped: hypothesis generation, with liability filters that cannot be switched off, and no objective that rewards toxicity. What was built is BRICS recombination of molecules the caller supplies. It is not a trained generative model. PAINS, Brenk, and NIH cannot be disabled. Requests that mention toxicity or hazard are rejected. Validity, uniqueness, novelty, diversity, and filter pass rate are calculated for each request. No fixed benchmark values for those metrics were written down. pytest 105 passed, 3 deselected, 0 failed. A frontend unit-test count was not recorded. Playwright was not re-run. Branch `m6e-generation`.

6F · Workflows

The pack asks for a declarative workflow over the services already built, resume from the last successful step, and a skip with a reason when a tool is missing. LLM steps stay off unless an environment variable enables them. Branch `m6f-workflows` accepts JSON. Steps are structure_overview, compound_descriptors, or docking. An end-to-end run on the synthetic peptide and ethanol returned a polymer-residue count and canonical SMILES CCO. The residue count from that run was not written down. With Vina hidden, docking and the dependent step were skipped. LLM summarize returned HTTP 503 with the flag off. With `BEAI_LLM_ENABLED` on, [PMID:9] was rejected when only PMID:1 was retrieved. pytest 109 passed, 3 deselected (network and model), 0 failed. Playwright was not re-run.

Plan-gaps close

Branch `plan-gaps`, from `m6f-workflows`, records the work the earlier sessions had left open: the 10-complex redocking table, DailyMed, Europe PMC, and Open Targets fixtures, an OpenMM checkpoint resume check, and a workflow docking step that launches Vina. Separate commands in the status note: `test_safety.py` 12 passed, 0 failed; the checkpoint test 1 passed; workflow tests plus one docking test 6 passed, 0 failed. The full suite then was pytest 118 passed, 3 deselected (network and model), 0 failed. Vitest 9 passed, 0 failed. The frontend build completed. fpocket is still absent. The NVE bound of 0.02 was left unmet. Playwright was not re-run. Source: docs/remaining-plan-status.md. The same 118 / 3 / 0 suite is in AGENTS.md.

Recorded measurements

The figures use only numbers printed in the cited files. A missing bar is a missing result. It is not zero. Where a label on a chart is shorter than the source float, the caption says so and the table keeps the recorded value.

Structure counts

1UBQ and 4HHB were downloaded from files.rcsb.org and checked against an RCSB Data API snapshot taken 2026-10-11. Chain A of 1UBQ has 76 polymer residues in both PDB and mmCIF. 4HHB observed polymer counts are A 141, B 146, C 141, D 146 (total 574). Water residue counts are 58 for 1UBQ and 221 for 4HHB, matched to `deposited_solvent_atom_count`. 1UBQ atom count is 660 and hydrogen count is 0. R-free is null because the file records it as missing. HEM is on 4HHB chains A–D (4 residues). PO4 is on chains B and D (2 residues). Those ligand residue counts are in the validation narrative. The snapshot names the component ids and chains and does not store the integers 4 and 2.

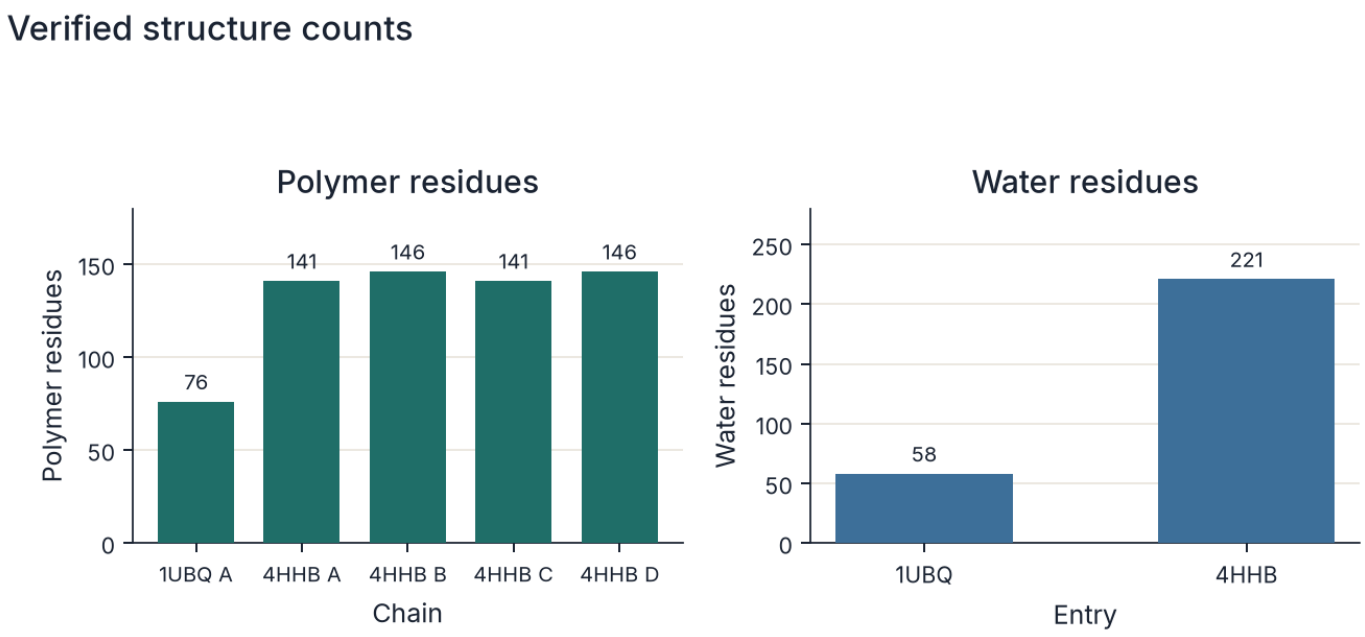
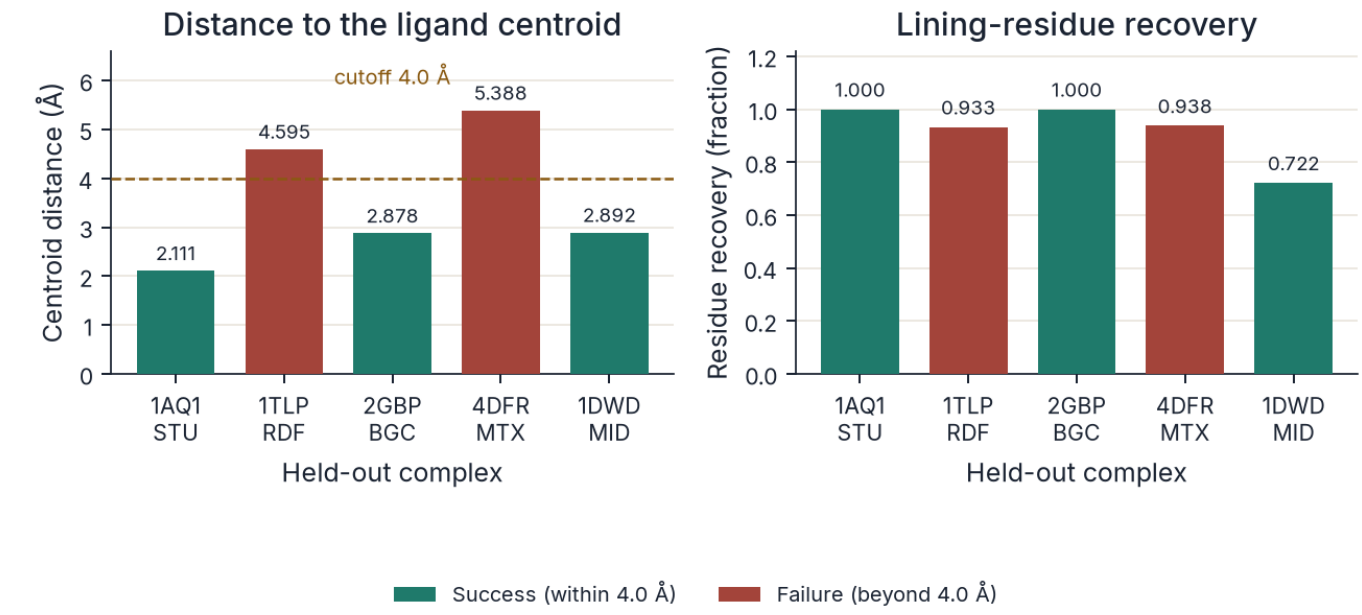


Figure 1. Polymer residues by chain, and water residues by entry. Both panels start at zero. 1UBQ has one polymer chain in this check.

Source: `docs/SCIENTIFIC_VALIDATION.md`, *Structure overview table, passed 2026-10-11. Same counts in* `backend/tests/fixtures/rcsb_entry_snapshot.json` *and in* `docs/milestone-1-status.md`.

Pocket detector, held-out

Success was pre-registered as a top-3 pocket centroid within 4.0 Å of the ligand heavy-atom centroid. The ligand is the largest non-buffer HET residue. Recovery is the fraction of polymer residues within 4.0 Å of that ligand that also line the closest top-3 pocket. Held-out success was 3 of 5. The validation file says that narrow claim is not a general binding-site validation. fpocket was not on PATH and did not contribute. Tuning, in the same table and not plotted here, was also 3 of 5: 3PTB 0.565 Å (recovery 1.000), 1STP 2.245 Å (0.750), 1HSG 0.675 Å (1.000), 1FKF 5.460 Å (0.643), 2PK4 12.835 Å (0.000). Failures in the full table are 1FKF, 2PK4, 1TLP, and 4DFR.



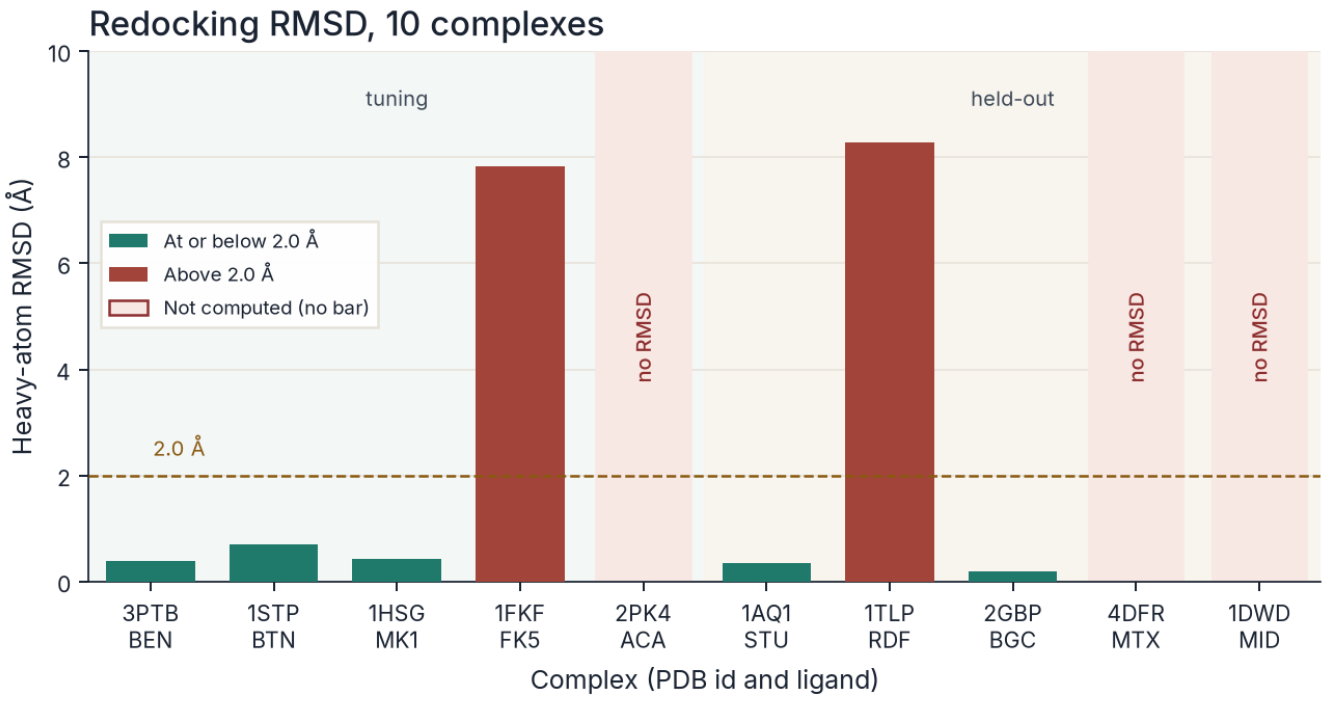
Source: docs/SCIENTIFIC_VALIDATION.md, Pocket detector table, held-out rows. Run 2026-10-13. Bar labels are the values printed in that table.

Figure 2. Held-out pocket results for five complexes. The dashed line is the pre-registered 4.0 Å cutoff. Recovery uses the same color as the distance result so a miss stays visible when recovery is high. 1TLP recovery is 0.933 and 4DFR recovery is 0.938, and both missed the cutoff.

Source: `docs/SCIENTIFIC_VALIDATION.md`, *Pocket detector table, held-out rows. Files downloaded 2026-10-13. Parameters are in that section (grid 1.0 Å, protein radius 2.5 Å, shell 5.0 Å, at least 5 of 7 directions, clusters of at least 8 voxels).*

Redocking RMSD

Ten complexes, the same split as the pocket benchmark, were redocked on 2026-10-13 with AutoDock Vina 1.2.5, exhaustiveness 8, seed 1, one mode, and a 20 Å box on the crystal ligand heavy-atom centroid. Heavy-atom RMSD uses RDKit `CalcRMS` with symmetry, in the crystal frame, without superimposing the pose onto the crystal ligand. Success is at or below 2.0 Å. Tuning success was 3 of 5. Held-out success was 2 of 5. 2PK4, 4DFR, and 1DWD failed receptor preparation and have no pose and no RMSD. 1FKF and 1TLP docked and missed 2.0 Å. The score column is a docking score, not an affinity. A separate validation row, score versus activity, says not run. The Seeds row in that same checklist also says not run. The 0.006 kcal/mol figure above is the 3PTB seed-1 pair only.



Source: docs/SCIENTIFIC_VALIDATION.md, Redocking RMSD table. Command: .venv/bin/python -m backend.app.services.docking.redock on 2026-10-13. Bar heights are the full recorded floats. A missing bar is an absent RMSD, not 0 Å. Repeated in the store note remaining-plan-status.md.

Figure 3. Heavy-atom RMSD for every complex that produced a pose. Pale red columns are preparation failures. They have no bar, because no RMSD was computed. The dashed line is the 2.0 Å success cutoff.

Source: `docs/SCIENTIFIC_VALIDATION.md`, *Redocking RMSD table. Command recorded there:* `.venv/bin/python -m backend.app.services.docking.redock` *on 2026-10-13. The plan-gaps status note repeats the success counts and the three failures with no RMSD.*

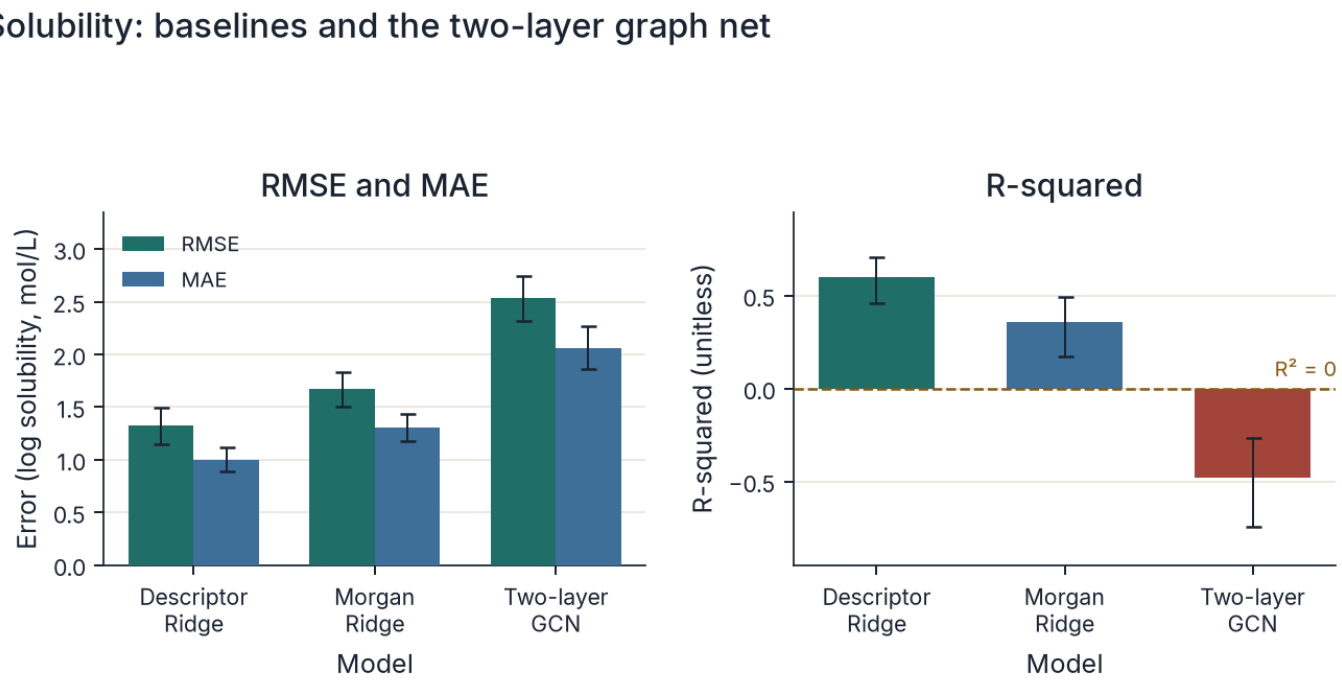
Preparation notes from the same section: 2PK4 failed in Meeko with excess inter-residue bond(s) at A:32. 4DFR failed the same way at B:131. 1DWD failed with CGLN heavy_miss=9, heavy_excess=0, and empty bond miss and bond excess sets. The table uses the full RMSD strings from the source. Scores are copied as printed, including the mixed number of decimals.

Split	PDB	Ligand	Heavy-atom RMSD (Å)	Success	Vina score (kcal/mol)
tuning	3PTB	BEN	0.38634036044964726	yes	-6.196
tuning	1STP	BTN	0.6946867472014318	yes	-7.04
tuning	1HSG	MK1	0.42136429418226423	yes	-11.567
tuning	1FKF	FK5	7.823345721495985	no	-6.548
tuning	2PK4	ACA	not computed	no	not computed
held-out	1AQ1	STU	0.3478932885966152	yes	-13.841
held-out	1TLP	RDF	8.278915104289972	no	-8.39
held-out	2GBP	BGC	0.19162198907988706	yes	-7.357
held-out	4DFR	MTX	not computed	no	not computed
held-out	1DWD	MID	not computed	no	not computed

Table. The ten-complex redocking record. Rows with no RMSD are shaded. Source: docs/SCIENTIFIC_VALIDATION.md.

Solubility baselines and the graph net

Dataset: `delaney-processed.csv` from the URL named in DeepChem's `load_delaney`, retrieved 2026-10-13. sha256 `8c06a76f0c6487d29ab0f903e6a7a7139f189ab3c1178f159c8be8964602f189`, 96,699 bytes, 1,128 data rows. The CSV has no licence line. The target column is "measured log solubility in mols per litre". The ESOL predicted column was not used as a feature or as a prediction. Morgan fingerprints are radius 2 and 2,048 bits. A second run with the same seed matched RMSE within 1e-6. None of these numbers are experimental solubilities. Applicability uses a pre-registered maximum Morgan Tanimoto of 0.3 against the training set. That threshold was not fit to the test labels.



Source: docs/SCIENTIFIC_VALIDATION.md, Solubility (session 6C). Error bars are the recorded 2.5 and 97.5 bootstrap percentiles (1000 resamples, seed 20261013), not a standard error. Point estimates and intervals are the full floats in that table. Same point estimates in the store note remaining-plan-status.md.

Figure 4. Held-out scaffold split (226 test rows). Teal and blue bars on the left are RMSE and MAE. On the right, the two-layer GCN is below zero: R-squared −0.476 on this split. The dashed line is $R^2 = 0$, the mean of the test labels, which is the comparison the validation file states. Error bars are bootstrap percentiles, not a laboratory uncertainty.

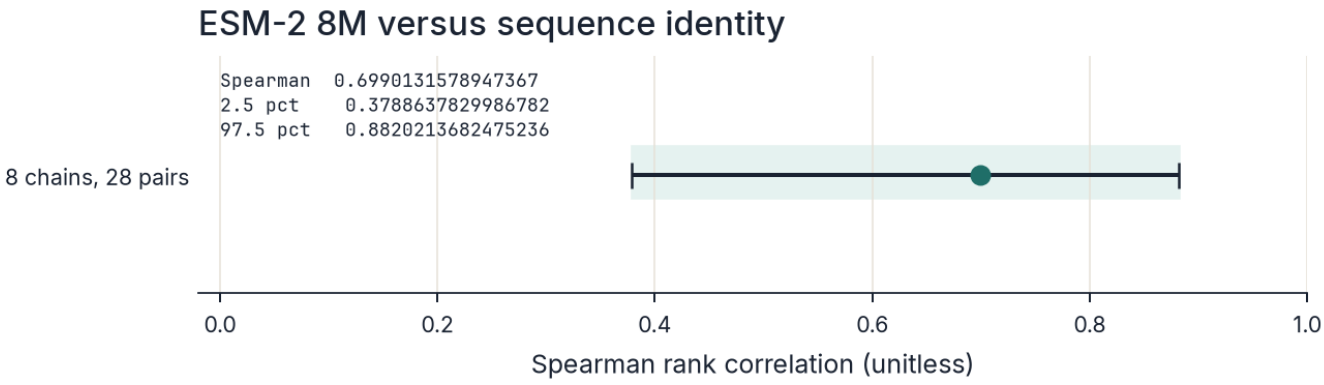
Source: `docs/SCIENTIFIC_VALIDATION.md`, *Solubility (session 6C) table. Point estimates also appear in* `docs/remaining-plan-status.md`.

Model	Metric	Estimate	2.5 percentile	97.5 percentile
Descriptor Ridge	RMSE	1.3217993772506413	1.147560194604866	1.4920541722551173
Descriptor Ridge	MAE	1.0004937427916745	0.8907326010644745	1.1141110380395272
Descriptor Ridge	R-squared	0.5985128816471968	0.45973007459714554	0.7074329222377606
Morgan Ridge	RMSE	1.6696791869919676	1.506008911536875	1.8300778122281194
Morgan Ridge	MAE	1.3029135097542737	1.1760270767204246	1.4366200664112465
Morgan Ridge	R-squared	0.3593709962979724	0.1757942096236605	0.4941057951191802
Two-layer GCN	RMSE	2.5347372633569814	2.3179658213153536	2.7474698037745178
Two-layer GCN	MAE	2.056036578614957	1.8600800380376539	2.264292208255532
Two-layer GCN	R-squared	-0.4764081172372039	-0.7456244760034777	-0.2633770633818176

Table. Full recorded solubility metrics. Each float is the value printed in the source. The GCN rows are shaded. The R-squared on that model is negative. Source: docs/SCIENTIFIC_VALIDATION.md.

ESM-2 Spearman

Kept chains, written down before the coefficient was computed: 1UBQ A 76, 4HHB A 141, 4HHB B 146, 3PTB A 223, 1STP A 159, 2GBP A 309, 1HSG A 99, 1HSG B 99. Eight sequences, 28 pairs. Identity is `identity_including_gaps` from a Biopython global alignment, BLOSUM62, open gap −10, extend −1. Embedding cosine uses the mean of the last hidden state over residue tokens. The interval is the 2.5 and 97.5 percentiles of 1,000 resamples, seed 20261013. The 1,000 individual coefficients were not recorded. All 1,000 resamples were finite.



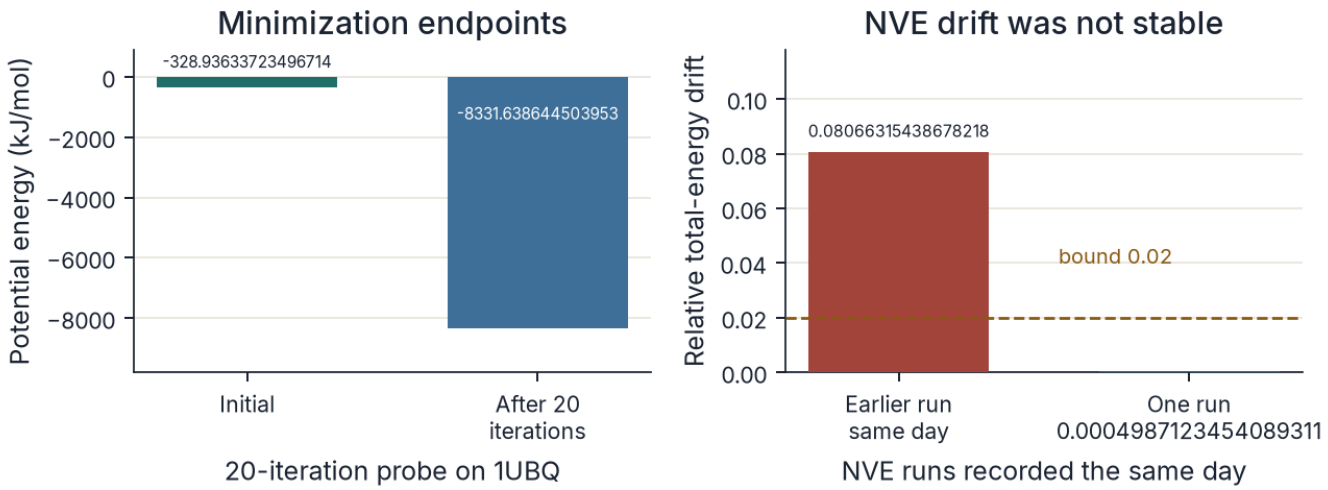
Source: docs/SCIENTIFIC_VALIDATION.md, Embeddings (session 6B). The marker and the bar are the recorded Spearman and the 2.5 and 97.5 percentiles of 1000 resamples (seed 20261013). Individual resample coefficients were not recorded. All 1000 resamples were finite. Same numbers in the store note remaining-plan-status.md.

Figure 5. One Spearman coefficient and its recorded percentile interval. The shaded band is that interval. There is no series of pairs in the document, so none is plotted. The annotation prints the full recorded Spearman and both percentiles.
Source: `docs/SCIENTIFIC_VALIDATION.md`*, Embeddings (session 6B). Spearman 0.6990131578947367. 2.5 percentile 0.3788637829986782. 97.5 percentile 0.8820213682475236. Command: pytest with* `-m model` *on* `backend/tests/test_embeddings_model.py`*, 1 passed.*

OpenMM minimization and NVE drift

OpenMM 8.6.1 on this machine listed Reference and CPU. Preparation is PDBFixer 1.12.0 at pH 7.0, force fields `amber14-all.xml` and `implicit/gbn2.xml`, nonbonded method NoCutoff, constraints HBonds, CPU platform with `OPENMM_CPU_THREADS=1`. The automated minimization test uses 5 iterations and asserts that the final potential energy is below the initial. Those two energies were not recorded. A separate 20-iteration probe the same day is the pair plotted here. NVE was 20 steps, 1 fs, Verlet, zero initial velocities, after 20 minimization iterations. The pre-registered relative total-energy drift bound was 0.02. One run was under it and an earlier run the same day was over it. The validation file says the bound is not treated as met.

OpenMM on 1UBQ



Source: docs/SCIENTIFIC_VALIDATION.md, OpenMM (session 6D). Only these endpoints were recorded. The 0.02 line is the pre-registered relative-drift bound. It was not stable, and the document says the bound is not treated as met. Drift labels are the full recorded ratios. Potential-energy endpoints are in the PDF table, not shortened on the bars.

Figure 6. Left: the two recorded potential energies of the 20-iteration minimization probe. The bars use the full recorded energies. Intermediate steps were not recorded, so there is no trajectory. Right: the two recorded NVE relative drifts on a linear axis. The earlier run is above the dashed bound. The other run is drawn and labeled with its full recorded ratio because the bar is too short to see against that scale. The bound was not stable.

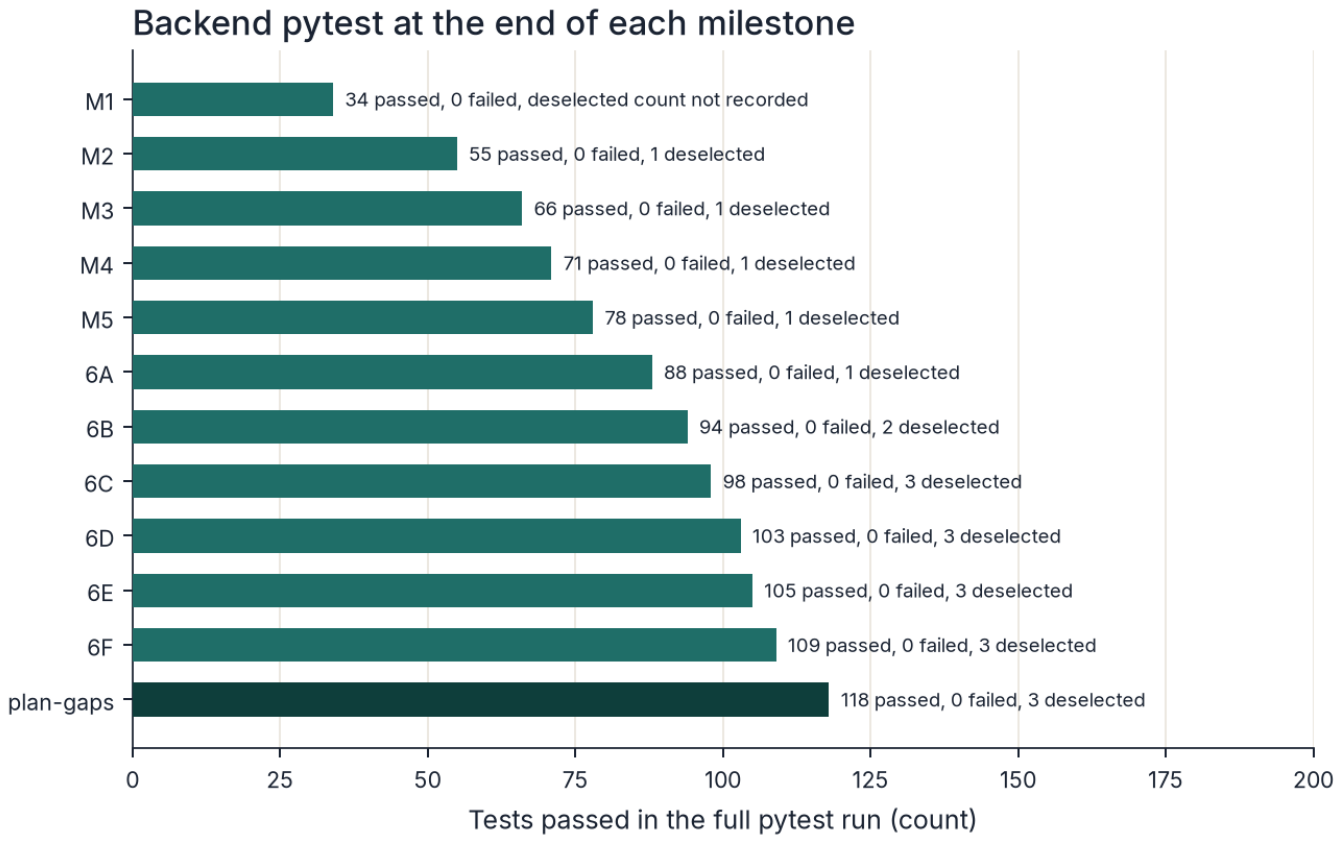
Source: `docs/SCIENTIFIC_VALIDATION.md`, *OpenMM (session 6D). The plan-gaps status note says this bound was left unmet and the recorded NVE numbers were not changed.*

Record	Value
20-iteration potential energy, initial	`-328.93633723496714 kJ/mol`
20-iteration potential energy, final	`-8331.638644503953 kJ/mol`
5-iteration potential energies	`not recorded; the test asserts final below initial`
NVE earlier run, relative drift	`0.08066315438678218`
NVE earlier run, E0	`-694.7754278681138 kJ/mol`
NVE earlier run, E1	`-750.818205470382 kJ/mol`
NVE one run, relative drift	`0.0004987123454089311`
NVE one run, E0	`-10170.949423790193 kJ/mol`
NVE one run, E1	`-10176.021801832367 kJ/mol`
Pre-registered NVE bound	`0.02, not treated as met`
Langevin mean kinetic temperature	`260.1331467073965 K (window 150 K to 450 K)`
Same-seed final total-energy gap	`1.6878309150943096 kJ/mol (bound 1.0 kJ/mol missed)`
Checkpoint versus a continuous run	`maximum absolute coordinate difference 7.993713069254227e-08 Å (bound 1e-5 Å)`

Table. OpenMM numbers as recorded. The shaded row is the unmet drift bound. Langevin was 15 steps, 1 fs, seed 7, after 15 minimization iterations. The checkpoint check reloaded one prepared PDB, because a second PDBFixer call does not reproduce coordinates. It did not change the NVE numbers. User dynamics are capped at 500 steps and minimization at 100 iterations. Source: docs/SCIENTIFIC_VALIDATION.md.

pytest counts

These are the full-suite counts in the status notes. Milestone 1's geometry section also records an earlier run of 31 passed, 0 failed, during the structure-overview tests. The bar uses the 34 passed, 0 failed figure from the milestone status, which is the completed Milestone 1 command. Deselected tests are the network marker, and from 6B also the model marker. Milestone 1's status line does not state a deselected count.



Source: docs/milestone-1-status.md for M1 (34 passed, 0 failed; deselected not stated). docs/remaining-plan-status.md for M2 through plan-gaps. AGENTS.md records the plan-gaps suite as 118 passed, 3 deselected, 0 failed (2026-10-13). Each bar is the full-suite pass count at that point, not the number of new tests.

Figure 7. Backend pytest passed counts. Every one of these runs recorded 0 failed. The count rises because the suite grew. It is not a score. plan-gaps is the latest recorded full run.
Source: `docs/milestone-1-status.md` *(Milestone 1) and* `docs/remaining-plan-status.md` *(Milestone 2 through plan-gaps).* `AGENTS.md` *records the plan-gaps result as 118 passed, 3 deselected, 0 failed.*

Frontend Vitest, where the status notes state it: Milestone 1, 2, and 3 each 7 passed, 0 failed. Sessions 6A, 6B, 6C, and 6D, and the plan-gaps close, each 9 passed, 0 failed. Vitest counts for Milestones 4 and 5 and sessions 6E and 6F were not in those notes. Playwright: 1 passed, 0 failed on Milestone 1. It was not re-run after that.

Other checks that were written down

These are single checks or short code lists. They are not plotted, because the figures above are the series the summary was asked to graph. All of them are in docs/SCIENTIFIC_VALIDATION.md.

Check	Recorded result
Analytic radius of gyration	Two equal points 3.0 Å apart give unweighted Rg 1.5 Å, tolerance 1e-6. Passed in a pytest run of 31 passed, 0 failed (2026-10-11).

Check	Recorded result
Ubiquitin average mass	Length 76. Biopython average mass 8564.736 Da. Independent elemental sum 8564.737 Da. Difference 0.00086 Da. Average mass, terminal water included.
1UBQ SASA	Biopython Shrake-Rupley 4893.04 Å². Biotite sasa 4834.72 Å². Probe 1.4 Å, 100 points, waters excluded. Relative difference 1.19%. Pre-registered tolerance 20% of the Biopython total.
1UBQ DSSP (mkdssp 4.2.2)	H 12, G 6, I 0, E 24, B 2, T 12, S 4, P 4, loop 12. File HELIX ranges 23–34 (class 1) and 56–59 (class 5). The validation file explains the overlap and says the counts are not expected to match the depositor records.
1UBQ self-superposition	C-alpha RMSD within 1e-6 Å, 76 pairs. A two-residue chain failed the job and named the 3-pair minimum.
Compound contract checks	Aspirin CC(=O)Oc1ccccc1C(=O)O, C9H8O4, 180.16. Caffeine C8H10N4O2, 194.19. Ethanol CCO, C2H6O, 46.07. RDKit 2025.09.1.
3PTB / BEN parser check	Modes −6.073, −5.062, −4.839 kcal/mol. Second seed-1 run differed by 0.006 kcal/mol. Window 0.05 kcal/mol.

Named, and not given a number

- Docking score versus activity, one target and one ChEMBL standard type: the validation checklist says not run.
- The docking Seeds checklist row (score reproduction on the same inputs, beyond the single 3PTB pair): not run.
- Five-iteration minimization energies: not recorded. The test records only that the final energy is below the initial.
- NVE energies at each step: not recorded. The two relative drifts and their endpoint total energies are the record.
- The 1,000 ESM-2 bootstrap coefficients: not recorded. The percentiles and the statement that all 1,000 were finite are the record.
- Generation validity, uniqueness, novelty, diversity, and filter pass rate: calculated per request. No fixed model-quality number is in the documents.
- The polymer-residue count from the session 6F peptide-and-ethanol workflow: described as a real count, integer not written down. Canonical SMILES CCO was written down.
- Vitest counts for Milestones 4 and 5 and sessions 6E and 6F: not in the status notes.
- Milestone 1 deselected count: not stated next to 34 passed, 0 failed.
- 4HHB atom count: not in the validation section or the RCSB snapshot used here. It is omitted.

Limits

These are limits the documents still record after the plan-gaps close. Earlier “not built yet” lists in the known-limitations file describe the milestone that was current when they were written.

fpocket is absent

fpocket was not on PATH for the pocket benchmark, and it is still absent. The plan-gaps note says `apt-cache search fpocket` returned no package, and the Ubuntu archive has no fpocket package. Pocket jobs skip that stage and say so. The baseline geometric detector still runs. Held-out success of 3 of 5 is the recorded claim, and the known-limitations file says the detector is not a general binding-site predictor.

The NVE drift bound was not stable

The pre-registered relative total-energy drift of 0.02 was not stable on 1UBQ. The earlier run the same day was 0.08066315438678218. Another run was 0.0004987123454089311. The validation file says the 0.02 bound is not treated as met. Plan-gaps left that result in place. On a separate check, two Langevin runs on the same prepared system differed by 1.6878309150943096 kJ/mol in final total energy, and the pre-registered 1.0 kJ/mol reproduction bound was missed. A short simulation is not evidence of stability, binding, or efficacy. Minimization is not molecular dynamics.

Playwright was not re-run after Milestone 1

The browser smoke test passed once: 1 passed, 0 failed, with both servers running, on Milestone 1. AGENTS.md says it was not re-run after Milestone 2 panels. The known-limitations notes for sessions 6E and 6F, and the plan-gaps status, say Playwright was not re-run. The unpacked Chromium cache was about 681 MB (zips about 191 MB, 120 MB, and 2.2 MB). Firefox and WebKit were not installed.

Optional tools

The kickoff audit (2026-10-11) found no project virtualenv, and system imports of RDKit, Biopython, Biotite, OpenMM, and PyTorch failed. mkdssp, vina, obabel, fpocket, pytest, uv, and conda were absent. `nvidia-smi` was absent. RAM was 15 GiB total, about 4.9 GiB available, no swap. Milestone 1's health check still reported RDKit, mkdssp, Vina, OpenMM, PyTorch, and GPU unavailable.

Later documents record tools that were installed after that: mkdssp 4.2.2, Biotite 1.6.0, RDKit 2025.09.1, AutoDock Vina 1.2.5, torch 2.9.1+cpu, transformers 5.19.0, OpenMM 8.6.1, and PDBFixer 1.12.0. OpenMM's platform list on this machine remained Reference and CPU. No later document records a GPU, or CUDA, OpenCL, or Metal.

Still recorded as unavailable at the plan-gaps close: fpocket. obabel is listed as absent from PATH in the Environment section of docs/KNOWN_LIMITATIONS.md, and later notes do not record installing it. That same Environment section still says OpenMM and PyTorch are not importable and that vina is absent from PATH. Those three sentences disagree with the later session notes and with docs/SCIENTIFIC_VALIDATION.md, which record the OpenMM, torch, and Vina runs plotted here. This summary follows the later runs for those three tools.

Other limits the files state in force

- Relative SASA is not reported. No maximum-accessibility table is cited.
- Hydrogen bonds are backbone heavy-atom estimates. 1UBQ has no hydrogens.
- Radius of gyration in the overview is unweighted over C-alpha atoms.
- Chain breaks are C-alpha distances above 4.5 Å by default. They are not crystallographic annotations.
- Sequence mass is an average mass with terminal water included. It is not an experimental mass.
- Embeddings are not experimental measurements and are not evidence of function or structure. Sequences longer than 1026 tokens, including special tokens, are refused.
- Solubility outputs are machine-learning predictions. The Delaney CSV is not committed. The file itself has no licence line.
- A docking score is not an affinity. Score versus activity was not run.
- Safety evidence has no accuracy target. The pages are research evidence review, not dosing or medication advice.
- Generation outputs are computational hypotheses: unsynthesized and unvalidated. Filters cannot be disabled.
- A workflow does not draw a scientific conclusion. Step outputs keep their own origins. LLM summarize is off unless the environment flag is set.
- There is no CI workflow recorded in the known-limitations note. Jobs are an in-process thread pool.
- PDBFixer removes heterogens and adds atoms. That changes the input structure.
- The README quick start is the command sequence used to create the environment. A second clone into an empty directory was not performed.

Sources

Every number in this PDF comes from one of these files. Paths are absolute so a reader outside the checkout can find the status notes. Repository paths are relative to the plan-gaps working tree.

`/workspace/README.md`
What the running app does, ports 8731 and 8732, the optional-tool note, and the Playwright cache size.

`/workspace/AGENTS.md`
The platform definition, the AnalysisResult rule, and the plan-gaps test line: pytest 118 passed, 3 deselected, 0 failed; Vitest 9 passed; Playwright 1 passed on Milestone 1 and not re-run after Milestone 2.

`/workspace/docs/SCIENTIFIC_VALIDATION.md`
Structure counts, pocket table, redocking table, compound checks, ESM-2 Spearman, solubility metrics, and OpenMM minimization, NVE, Langevin, and checkpoint numbers. This is the source for Figures 1–6 and both data tables.

`/workspace/docs/ARCHITECTURE.md`
Stack, evidence-category meanings, and package versions cited in the milestone notes (RDKit, OpenMM, PDBFixer, torch).

`/workspace/docs/KNOWN_LIMITATIONS.md`
fpocket, the unmet NVE bound, Playwright not re-run, missing GPU platforms, and the other limits in the last section. The Environment section is partly older than the later session headings; that conflict is stated above.

`/workspace/docs/ROADMAP.md`
Milestone order and aims as written at kickoff (the file's opening line says the codebase had no application on 2026-10-11). Later status notes are the record of what was built.

`/workspace/docs/AUDIT.md`
Kickoff inspection on 2026-10-11: empty main tree, tool absence, RAM, and no GPU. Those facts are labeled as the audit, not as the state after later installs.

`/workspace/backend/tests/fixtures/rcsb_entry_snapshot.json`
RCSB Data API fields retrieved 2026-10-11 that match the 1UBQ and 4HHB counts in Figure 1. Not a source of extra plotted numbers.

`/cursor/stores/bc-81995169-412c-4238-9aa7-9087f1bb41d9/docs/milestone-1-status.md`
Milestone 1 commands: pytest 34 passed, 0 failed; Vitest 7; Playwright 1. Structure counts repeated from the validation file.

`/cursor/stores/bc-81995169-412c-4238-9aa7-9087f1bb41d9/docs/remaining-plan-status.md`
Commands and pytest counts for Milestones 2–5, sessions 6A–6F, and the plan-gaps close. Figure 7. Also the ESM-2, solubility, and NVE point estimates, which match the validation file.

`/cursor/stores/bc-81995169-412c-4238-9aa7-9087f1bb41d9/docs/kickoff-status.md`
Kickoff verification that main was empty, the machine had no project virtualenv, and residue counts had not been taken yet. No plotted number comes from this file.

`/cursor/stores/bc-81995169-412c-4238-9aa7-9087f1bb41d9/internal/beai-cursor-prompt-pack.pdf`
Priority order (scientific correctness, working software, reproducibility, modularity, honest uncertainty) and the aim of each milestone prompt. 11 October 2026.

Figure files, kept next to this summary: media/structure-counts.png (Figure 1), media/pocket-held-out.png (Figure 2), media/redock-rmsd.png (Figure 3), media/solubility-metrics.png (Figure 4), media/esm2-spearman.png (Figure 5), media/openmm-minimization-nve.png (Figure 6), media/pytest-by-milestone.png (Figure 7).

Research software. Not a clinical tool. No medical advice.